

ST ANTHONY, NL, Canada

WMO: 718300

Lat: 51.390N

Lon: 56.090W

Elev: 33

StdP: 100.93

Time Zone: -3.50 (NAF)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-22.5	-20.5	-26.8	0.3	-21.8	-24.6	0.4	-20.2	18.7	-6.3	17.0	-6.9	5.9	270	0.886

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.9	23.1	17.0	21.6	16.5	20.2	15.7	18.4	21.5	17.4	20.4	16.4	19.3	8.1	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	12.4	20.0	16.1	11.5	19.0	15.0	10.7	18.1	52.4	21.6	49.1	20.5	46.2	19.4	22.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.0	13.9	12.6	-26.6	26.3	2.0	1.1	-28.0	27.0	-29.1	27.7	-30.2	28.3	-31.7	29.1
			-26.9	20.6	1.9	0.8	-28.2	21.2	-29.4	21.7	-30.4	22.1	-31.8	22.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.2	-9.6	-9.8	-6.0	-0.9	4.2	9.5	14.2	14.7	10.6	5.0	-0.3	-5.7	(6)
	DBStd	9.47	5.83	5.76	5.16	3.25	2.98	3.63	3.35	2.93	3.54	3.27	4.03	5.31	(7)
	HDD10.0	3217	609	553	497	326	182	54	6	2	34	159	309	486	(8)
	HDD18.3	5887	867	787	756	576	438	266	132	116	233	414	558	744	(9)
	CDD10.0	381	0	0	0	0	2	38	136	148	52	4	1	0	(10)
	CDD18.3	8	0	0	0	0	0	0	4	4	1	0	0	0	(11)
	CDH23.3	35	0	0	0	0	0	3	19	12	1	0	0	0	(12)
	CDH26.7	1	0	0	0	0	0	0	0	1	0	0	0	0	(13)
(14) Wind	WSAvg	6.1	6.6	6.8	7.0	6.6	5.7	5.6	5.3	5.2	5.9	5.8	6.1	6.6	(14)
(15) Precipitation	PrecAvg	1099	111	92	78	60	69	81	100	98	95	94	89	104	(15)
	PrecMax	1315	342	151	143	126	158	178	201	217	217	162	146	166	(16)
	PrecMin	833	53	31	20	12	16	37	39	17	39	25	37	50	(17)
	PrecStd	128	63	32	28	28	37	36	39	40	35	34	28	30	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.4	3.8	3.9	9.2	17.4	22.9	25.4	24.7	22.2	15.9	13.3	6.0	(19)
		MCWB	4.0	2.8	2.5	5.5	11.6	15.5	18.2	17.9	17.6	12.4	11.7	5.4	(20)
	2%	DB	1.7	1.6	2.3	5.8	14.0	20.2	23.4	22.9	19.5	13.5	9.2	3.9	(21)
		MCWB	1.2	0.8	1.0	3.8	9.8	14.2	17.2	17.2	15.8	11.3	8.2	3.4	(22)
	5%	DB	0.3	0.4	1.3	4.3	11.7	18.2	21.9	21.5	17.9	11.7	6.7	2.4	(23)
		MCWB	-0.2	-0.3	0.4	2.6	8.2	13.1	16.7	16.7	14.7	9.9	5.8	1.9	(24)
	10%	DB	-1.0	-0.9	0.4	3.0	9.5	16.2	20.3	20.1	16.2	10.0	5.1	1.1	(25)
		MCWB	-1.6	-1.5	-0.4	1.5	6.8	12.2	15.7	16.1	13.6	8.3	4.2	0.5	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.0	3.1	2.9	6.7	12.4	17.0	19.9	19.7	18.4	13.9	11.8	5.5	(27)
		MCDB	4.4	3.5	3.5	8.2	15.3	20.4	23.2	22.3	20.7	15.6	13.1	5.8	(28)
	2%	WB	1.3	1.0	1.3	4.2	10.4	15.3	18.5	18.5	16.6	11.7	8.3	3.5	(29)
		MCDB	1.6	1.5	1.9	5.3	13.4	18.8	21.8	21.4	18.9	13.0	9.0	3.8	(30)
	5%	WB	-0.1	-0.2	0.5	2.8	8.5	13.8	17.4	17.5	15.2	10.1	6.0	2.0	(31)
		MCDB	0.3	0.3	1.2	4.0	11.2	17.2	20.6	20.3	17.2	11.5	6.6	2.3	(32)
	10%	WB	-1.6	-1.5	-0.4	1.7	6.8	12.4	16.3	16.6	13.9	8.5	4.3	0.6	(33)
		MCDB	-1.1	-1.0	0.4	2.7	9.3	15.9	19.5	19.4	15.9	9.7	5.0	1.0	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	7.9	8.2	7.5	6.0	7.5	8.7	8.7	7.9	7.4	6.1	5.8	6.6	(35)
		MCDBR	7.5	7.4	6.9	7.3	12.0	12.5	11.4	10.2	9.2	8.7	7.5	6.4	(36)
	5% WB	MCWBR	7.5	7.3	6.5	5.9	8.3	7.7	6.7	6.1	6.5	7.0	7.3	6.4	(37)
		MCWBR	7.7	7.7	6.8	6.5	11.1	11.3	10.3	9.1	8.5	8.2	7.5	6.4	(38)
(40) Clear-Sky Solar Irradiance	taub		0.281	0.287	0.310	0.368	0.389	0.384	0.407	0.377	0.354	0.322	0.296	0.285	(40)
		taud	2.299	2.304	2.285	2.214	2.289	2.370	2.357	2.435	2.474	2.533	2.490	2.321	(41)
	Ebn at Noon		734	842	888	867	864	872	843	853	835	802	731	665	(42)
		Edh at Noon	64	84	104	127	124	116	116	102	88	68	55	55	(43)
(44) All-Sky Solar Radiation	RadAvg		0.82	1.59	2.74	3.94	4.85	5.05	4.95	4.20	3.10	1.84	0.98	0.64	(44)
	RadStd		0.11	0.25	0.39	0.53	0.47	0.45	0.40	0.28	0.23	0.15	0.09	0.08	(45)

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	+0.64	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)		N/A	N/A	+1.44	N/A	+0.37	+0.55	N/A	N/A	N/A	N/A

Nomenclature: See separate page